

# The Trial

Methodology & Operations Lead • CLARION-3

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15 DAYS • 51 STOPS • GRADE 12 • THE\_TRIAL

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## THE OPENING CARD

CLARION-3 tests a new drug against the usual care. 2,400 people at 31 hospitals have joined it. Nobody running the trial may know which arm is ahead. You are the methods and operations lead, and in fifteen days you build The Monitoring Board Pack. A board of outsiders reads it in three weeks and decides whether more people are asked to join.

## The rules we cannot change

### BEFORE THE DAY — GREET

#### **One floor, four functions, and none of them may tell the others everything**

The coordinating centre is one long floor: statistics at one end, the monitors in the middle, safety and the sites at the other. Feldman, the trial statistician, wants you known to as many of them as you can before the first interim, because the firewall means the person you have to ask is often the one person who cannot tell you why they are asking back.

### PLAN CARD 36W

Monday, 21 days before the board meets. CLARION-3 has entered 2,400 people at 31 hospitals, and the trial chair already wants it stopped. Today you write down what the trial promised before the first patient joined.

### BRIEFING 12W

Three weeks to the second interim, and the board pack starts today.

### WHAT YOU DECIDE 9W

Establish what CLARION-3 promised before it had any data.

## 1.1 Concealment is not the same as blinding

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · CHOICE

### SCENE 40W

Farida Diouf, the trial pharmacist, has a note from site 7. Their pharmacy diary lists kit numbers beside the arm each one contains, and the research nurse can see the next entry before she decides whether to enter a patient.

### GUIDE 65W

All four options are real costs of a readable diary. Ask of each whether it acts at the moment of entry, or afterwards. A nurse who can see the next slot can wait a day for a patient she thinks will do well. Nobody has to be dishonest for that to wreck the comparison. Concealment protects entry; blinding protects everything after, and they fail separately.

### ASK 24W

Site 7 has lost allocation concealment — the next arm is no longer hidden when a patient is entered. What has that cost CLARION-3?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

participants are assigned to arms by chance rather than by choice

### VERDICT 87W

Randomisation only buys balanced groups if nobody can see what is coming. A nurse who knows the next slot is the drug can wait a day for a patient she thinks will do well, or enter a frailer one when the next slot is standard care. Neither is dishonest and both destroy the comparison, because the arms now differ in who was entered as well as in what they were given. Concealment protects the moment of entry. Blinding protects everything after it, and the two fail separately.

**TAKEAWAY 17W**

If the next assignment can be seen in advance, chance is no longer the only thing choosing.

## 1.2 Why this outcome was chosen

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A PERSON · STRESS

### SCENE 49W

Kwame Ansah chairs the endpoint adjudication committee. He has the original design memo beside the current event counts. Four years ago the team compared several candidate outcomes before anyone had trial data. A journalist has now asked why the protocol committed to the composite rather than the cleanest alternative.

### GUIDE 63W

Four candidate primary endpoints, scored on two things the sponsor cares about. The slider is the event rate the trial actually gets, which nobody knows in advance — slide it to the pessimistic end and see which endpoint still answers the question there. The one that looks best at the expected rate is not necessarily the one that survives being wrong about it.

### ASK 15W

Which prespecified endpoint was the most robust design choice before the event rate was known?

### BEHIND THE BUTTON 170W

What a primary endpoint is. The one question the trial was built to answer, named before the first participant was randomised and used to set the sample size. It also fixes how the result will be reported: an absolute risk reduction in percentage points, with the relative figure beside it. Everything else on Ansah's screen is collected — worth having, worth reporting, and not what the trial promised.

Why naming it in advance matters. With four measurements per participant, somebody choosing afterwards has four chances to find a difference, and the reported result would be the best of four rather than the answer to one question. That is the whole of why the definition is four years old.

Why the event rate is the assumption that bites. Every endpoint's power depends on how often its event happens. A rare, hard endpoint is unimpeachable and may never accumulate enough events to say anything; a composite happens more often and answers a blurrier question. That trade is what the slider makes visible.

### TAKES AS READ

a trial collects more than one measurement from each participant

### VERDICT 116W

The important choice happened four years ago, before anyone knew the event rate or treatment result. The design team could stress-test candidate endpoints against plausible event rates, but once the trial began the primary endpoint was fixed. Cardiovascular death alone asks the crispest question and would need more events at the low end of the plausible range. The composite occurs more often and remains estimable across that range, at the cost of combining events of different clinical weight. Walk distance is measurable even when hard events are rare, but it answers a different clinical question. This board therefore reconstructs why the composite

was a defensible prespecified compromise; it does not let today's data choose yesterday's endpoint.

**TAKEAWAY 19W**

The primary endpoint is the one question the trial was built to answer; the rest are collected, not promised.

### 1.3 A trial ends at a number of events

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

#### SCENE 36W

Yemi Balogun, the trial chair, wants to know when CLARION-3 will be finished and has been told, twice, that nobody can say. Feldman writes three quantities on the whiteboard and asks him to estimate it himself.

#### GUIDE 72W

Five numbers, and three belong to other questions: the events adjudicated so far, the hospitals recruiting, and one of the two rates. Ask of each whether this schedule depends on it. In this event-driven design, the planned information is driven mainly by adjudicated events, while event-free follow-up still contributes time at risk. The duration is therefore a forecast from recruitment, follow-up and event accrual, not a date the sponsor can simply choose.

#### ASK 20W

CLARION-3 is an event-driven trial. Estimate how long it has to follow its participants to reach the 380 planned events.

#### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

#### TAKES AS READ

the trial compares how often an event happens in two arms · events accumulate over time, at some rate per participant per year

#### VERDICT 115W

In a time-to-event trial, participants who remain event-free still contribute time at risk, but the precision of the treatment comparison is driven strongly by how many adjudicated events occur. So the design fixes a number of events, and recruitment and follow-up run until that number arrives. How long that takes is not a decision anybody makes — it is the required events divided by the rate they arrive at, and that rate belongs to the participants rather than to the trial. A healthier-than-expected group is bad news for the schedule for exactly this reason, and it is why the interim looks are placed at fractions of the events rather than at fractions of the years.

#### TAKEAWAY 21W

In this event-driven design, precision is driven mainly by adjudicated events; calendar time only determines how quickly those events can accrue.

A trial's claims are the ones it made before it had data.

## Which number can the board trust?

### PLAN CARD 42W

Tuesday, 20 days out, and the pack carries two outcomes rather than one. A committee rules on each event, while the site doctor writes down the symptom score. Today you find out where each number is made, and what each is worth.

### BRIEFING 14W

Two outcomes, two ways of measuring, and only one of them has a committee.

### WHAT YOU DECIDE 13W

Find out where a number is made before deciding what it is worth.

## 2.1 A number can be precise and still be the wrong outcome

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · PROTOCOL

### SCENE 46W

Kwame Ansah has 4 CLARION-3 measurements on his table. One comes from a national register, one from blinded readers, one from a treating doctor, and one from a central laboratory. Maggie Doyle, who speaks for the participants, wants to know what can go wrong with each.

### GUIDE 61W

Four measurements and four weaknesses. Pair them by asking how much trial judgement each measurement passes through. A death in a national register goes through almost none. A borderline admission goes through a definition. A symptom score goes through a person who may believe something. And one of them can be measured perfectly and still stand in for the wrong thing.

### ASK 12W

Match each measurement to the weakness that can bend what it means.

### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

### TAKES AS READ

a measurement can be precise without being clinically important · a surrogate outcome is a measure used to stand in for a clinical outcome

### VERDICT 79W

Measurements can fail in different ways. A death in a national register needs little trial judgement. A borderline admission can move if readers apply the definition differently, so blinded adjudication protects it. A symptom score can move with the scorer's belief about treatment. A laboratory marker can be measured perfectly and still mislead. If it stands in for a clinical outcome, it is a surrogate. Changing the marker does not prove that admissions, symptoms or survival changed with it.

#### TAKEAWAY 20W

Before trusting a number, ask both how it was made and whether it measures the outcome people actually care about.

## 2.2 Which deviation changes the question?

SITE OPERATIONS · MONITORS' ROOM · A PERSON · CHOICE

### SCENE 35W

Lucia Marchetti, the lead monitor, has four things sites did this month that were not what the protocol says. None of them harmed anybody. Site 12 has asked which of them are worth the paperwork.

### GUIDE 62W

All four are real deviations and none harmed anybody. Ask of each whether it lands on both arms equally. A late blood test and a late form are untidy in both arms. A different pump delivers the same drug. Only one of these changes who is in the trial. It matters because those entries need not be spread evenly across the arms.

### ASK 14W

Which deviation most changes the population or treatment contrast CLARION-3 was designed to estimate?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

the protocol states what should happen at each visit

### VERDICT 101W

Every deviation is documented, but they threaten different parts of the question. Entering people who do not meet the eligibility criteria changes the population to which the randomized comparison applies. A late blood draw can matter if timing is related to outcome or differs by arm; a late form mainly threatens completeness; a different pump matters if it changes delivered treatment. None can simply be dismissed as "equal in both arms" without checking. Here the ineligible entries most directly alter the trial's target population, so they require the strongest immediate impact assessment while the other deviations are still recorded and reviewed.

#### TAKEAWAY 17W

A deviation is recorded for what it does to the comparison, not for how bad it looks.

## 2.3 The plan that was in force

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · SEQUENCE

### SCENE 38W

Nkechi Umeh, the regulatory lead, has four versions of the analysis plan, each dated. Feldman wants to know which One governs the interim in 20 days, and the sponsor's lawyer wants to know how anybody could prove it.

### GUIDE 65W

All four steps will happen, so ask what order makes the claim provable.

Prespecification is a claim about time: this analysis was chosen before anybody could see what it would give. The only support for that afterwards is a dated public record made before the data existed. Reverse any two of these and the arithmetic is still right about a question chosen after the fact.

### ASK 8W

Find what the trial promised about this analysis

### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

### TAKES AS READ

a trial's plan can be amended while it is running

### VERDICT 96W

Prespecification is a claim about time: this analysis was chosen before anybody could see what it would give. The only way to support that claim afterwards is a public record with dates on it, made before the data existed. That is why the plan is registered rather than filed, why the cut comes after the registration rather than before, and why the person who runs it is the one person already permitted to see the arms. Reverse any two of those steps and the analysis is still correct arithmetic about a question chosen after the fact.

### TAKEAWAY 14W

A result is worth more when the plan that produced it can be dated.

A measurement is worth what the procedure that made it is worth.

## Two liver cases

### PLAN CARD 39W

Wednesday, 19 days out. Site 4 and site 19 have each sent in a patient who went into hospital with the same odd liver picture, 11 days apart. Today you decide what gets reported, to whom, and how fast.

### BRIEFING 13W

Two sites have sent the same kind of event in the same fortnight.

### WHAT YOU DECIDE 13W

Decide what has to be reported today rather than at the next meeting.

### 3.1 The clock that is not the analysis clock

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · TRIAGE

#### SCENE 42W

The second admission arrived 18 minutes ago. 1 patient is still on the ward, and the 2 sites are 400 kilometres apart. Helena Vogt, who chairs the independent monitoring board, is reachable this afternoon. The next board meeting is 19 days away.

#### GUIDE 57W

Four options, and two clocks are running. One patient is still on the ward and the board meets in nineteen days. Ask of each option which clock it obeys. A serious unexpected event may change treatment tomorrow. Whether it also counts as an endpoint is a separate path, and the same event can travel both at once.

#### ASK 8W

What should happen to these two reports today?

#### BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

#### TAKES AS READ

serious harm during a trial is reported whether or not the drug is thought to have caused it · hard endpoints against judged ones, and blinded adjudication — taken as read

#### VERDICT 96W

Safety reporting and endpoint analysis run on different clocks. Serious adverse events are reported promptly to the sponsor, with follow-up information as it becomes available; the independent monitoring board can receive arm-specific summaries through the protected unblinded channel. Whether the same event later counts toward a trial endpoint is a separate blinded process. The urgent report does not itself prove causality and it does not automatically decide recruitment. Pausing, continuing, or changing treatment is a separate risk decision made from the

emerging safety picture. Waiting 19 days for the scheduled efficacy meeting would confuse those clocks.

**TAKEAWAY 13W**

A safety report does not wait for the analysis it might belong to.

### 3.2 Ten minutes on each case

SITE OPERATIONS · MONITORS' ROOM · A PERSON · VALUE

#### SCENE 38W

Lucia Marchetti has ten minutes with each site before the reports go to Vogt. One coordinator is at a bedside. Five useful questions are on her screen, but asking all of them would take longer than either call.

#### GUIDE 59W

Ten minutes with each site, five useful questions, and no time for all of them. Open each question and ask what its answer would change before the reports go to Vogt. One coordinator is at a bedside, so the call has to be short and worth having. Spend the ten minutes on what the reports cannot already tell you.

#### ASK 14W

What should Marchetti spend the ten minutes finding out before the safety reports leave?

#### BEHIND THE BUTTON 132W

Why some questions are wasted on a call. Anything the site has already entered in the system is in the reports. A call is the only way to get at what has not been written down: why a visit slipped, what the coordinator is worried about, whether a protocol step is being interpreted differently.

Why 'before the reports leave' is the constraint. Once the report is out, a correction is a deviation and a conversation. Something learned now changes what the report says; the same thing learned tomorrow changes what has to be explained.

What a coordinator at a bedside can actually do. Answer one or two things from memory. A call that asks five questions gets three rushed answers, and the rushed ones are the least reliable data in the trial.

#### TAKES AS READ

a safety pattern depends on whether cases share timing and clinical features · a VALUE board charges time for each piece of evidence · hard endpoints against judged ones, and blinded adjudication — taken as read

#### VERDICT 79W

Two reports become a signal only when their details line up in a way that could matter. Timing is the fastest discriminator here. A case three days after first exposure tells a different story from one eight months later. Matching intervals in two distant sites would strengthen the case for a shared cause. Different intervals would weaken it. Investigator opinions, recruitment totals and withdrawal status may still matter, but none can establish that temporal pattern before the reports leave.

#### TAKEAWAY 19W

When time is scarce, collect the evidence that can change the decision before collecting detail that only decorates it.

### 3.3 Which clock does this event run on?

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · PROTOCOL

#### SCENE 53W

Umeh has four events from this month and three different clocks on the wall. Site investigators report serious adverse events promptly to the sponsor. The sponsor then decides which reports meet the applicable criteria for expedited regulatory reporting. One protocol-defined hospitalization is also the trial's primary endpoint and follows its blinded adjudication path.

#### GUIDE 78W

Four events, four routes. Ask first whether the event is serious enough for prompt sponsor reporting. Then ask whether it is unexpected and reasonably suspected to be related to the drug — that is what can trigger expedited sponsor reporting to regulators under the applicable rules. A protocol-defined endpoint may have a specific reporting arrangement written in advance, but that does not make safety obligations disappear. Non-serious events still stay in the trial record and periodic safety picture.

#### ASK 16W

Route each event through the correct safety or endpoint pathway without confusing seriousness, causality and timing.

#### BEHIND THE BUTTON 144W

Why "serious" and "related" are different questions. A road-traffic death is serious even when nobody believes the investigational drug caused it. A liver hospitalization may be serious, unexpected and plausibly drug-related. The investigator reports the serious event promptly; the sponsor's causality and expectedness assessment determines whether an expedited suspected-adverse- reaction report is also required.

Why endpoints need their own lane. Trials sometimes specify that events already captured as efficacy or safety endpoints follow a protocol-defined reporting process rather than duplicate immediate reports in exactly the same form. They are still counted, adjudicated and included in safety surveillance. "It is an endpoint" is not permission to make it invisible to safety review.

Why the mild rash stays in the record. Non-serious does not mean nonexistent. It is captured and contributes to the accumulating safety profile even when it does not trigger the fastest reporting clock.

#### TAKES AS READ

the protocol and applicable rules define which serious events require immediate investigator-to-sponsor reporting and which sponsor reports are expedited externally

- safety monitoring: adverse events, seriousness and expedited reporting — taken as read

#### VERDICT 107W

The clock is not set by how dramatic an event sounds. Investigators promptly report serious adverse events to the sponsor under the trial's rules. The sponsor then

evaluates expectedness and reasonable causal possibility to determine whether an expedited suspected-adverse-reaction report is required. A protocol-defined endpoint can have a special reporting process written in advance, but it still contributes to safety surveillance. The mild rash does not trigger the serious-event clock and it still belongs in the record. Separating these routes prevents two opposite errors: treating every serious event as proof of drug harm, and treating an efficacy endpoint as though safety no longer needs to see it.

**TAKEAWAY 23W**

Seriousness sets one reporting obligation; expectedness and suspected causality determine another. Endpoint adjudication is a separate path, not a substitute for safety surveillance.

### 3.4 Concealment is not the same as blinding — Review

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · CALLBACK · SEQUENCE

#### SCENE 43W

Farida Diouf, the trial pharmacist, is on a call with site 22, which has been entering participants from a drawer of sealed envelopes since March. Diouf wants their next entries done on the central allocation server, and the site coordinator wants the steps.

#### GUIDE 60W

Four steps that all happen inside a minute, so the rail is not a clock. Order them by what each one closes off. Ask of each whether the site could still change its mind about this patient afterwards, and whether anything about the arm has been given away yet. Those two answers are what the order has to keep apart.

#### ASK 15W

Put site 22's four entry steps in an order that keeps the next slot hidden

#### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

#### TAKES AS READ

participants are assigned to arms by chance rather than by choice

#### VERDICT 88W

Concealment is about what a site can know while it can still act. Envelopes in a drawer can be counted, held to the light, or taken out of order, and a coordinator doing any of that is choosing patients rather than accepting them. A central server closes the gap by ordering the steps: eligibility and consent are recorded, the entry is committed, and only then does a kit number come back. The number names a box rather than an arm, so the blind survives the entry as well.

#### TAKEAWAY 18W

Concealment is a property of the order the steps run in, not of how the envelope is stored.

Safety reporting runs on a different clock from the analysis.

## The peek that changes the trial

### BEFORE THE DAY — FOLLOW

#### **A monitoring visit, walked once at the pace it happens**

Source data verification is not reading a file — it is standing beside somebody while they find the original, and the visit has a rhythm. Stay with the monitor. The pauses are where the discrepancies get noticed, and somebody hurrying the visit finds nothing.

### PLAN CARD 40W

Thursday, 18 days out. The sponsor's medical director has asked in writing for the results split by arm, before the board meets. Today you work out what an extra look costs the trial, and who is allowed to take one.

### BRIEFING 14W

Somebody senior would like a peek at the numbers before the board sees them.

### WHAT YOU DECIDE 12W

Establish what looking early costs and who is allowed to do it.

## 4.1 What an extra look spends

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

### SCENE 38W

The medical director wants one unplanned endpoint summary before noon and has already drafted the email asking for it. Feldman has the two planned looks on the whiteboard and an empty box where this new request would sit.

### GUIDE 61W

Add up what the planned looks already spend, and compare it with what the trial has to spend in total. Each look has a boundary, and the error rate it uses is the price of that look. Pick the three planned amounts. The unplanned look's price is on the board too, and it is what the sum has no room for.

### ASK 12W

What do the planned looks already spend of the trial's error budget?

### BEHIND THE BUTTON 167W

What is being spent. A five per cent error rate is a promise about the whole trial, not about any one test. Every look at the accumulated data is another chance to catch a wandering difference at its most flattering moment, so each look consumes part of that budget and the budget does not refill.

How planned looks are paid for. The boundaries at the planned looks are made harder — an interim result has to be more extreme than the final one to stop the trial — and the amounts add up to no more than the total. That is the whole of the arithmetic: the spending plan is written before any data arrive.

Why an unplanned look is different in kind. It has no boundary and no allowance, so whatever it spends comes out of the final analysis. A trial that has spent more than its budget can no longer say what its error rate is, which is a worse position than not having looked.

### TAKES AS READ

a trial accepts a small chance of calling a difference real when there is none · type I and type II error, and what  $\alpha$  and  $\beta$  are for — taken as read

### VERDICT 132W

The trial has allocated its one-sided Type I error across the planned analyses:  $0.001 + 0.004 + 0.020 = 0.025$ . Those are authored incremental spends, not universal prices for looking. An extra efficacy analysis does not come with a fixed automatic cost such as 0.015; its effect on the overall false-positive rate depends on the new boundary and on the correlation between sequential looks. If an extra formal look is scientifically necessary, the boundaries have to be revised with a valid group-sequential method so the overall error remains 0.025. What cannot happen is an unadjusted peek followed by pretending the original operating characteristics are unchanged. There is a second cost too: sponsor access to arm-specific interim efficacy data can change how the trial is managed even if nobody performs a formal test.

**TAKEAWAY 14W**

A trial's tolerance for error is spent by testing, and it does not refill.

## 4.2 Why the board looks and the team does not

SAFETY MONITORING · MONITORING BOARD ROOM · A PERSON · CHOICE

### SCENE 41W

Vogt is in the building for an hour. The closed report kept by Alina Petrescu, the unblinded statistician, is on the table, face down. The medical director is copied on everything the trial team writes and on nothing the board writes.

### GUIDE 54W

Four options, and somebody has to see the arms or a harmful treatment continues unnoticed. Ask of each whether it names a route from knowledge into the trial's own conduct. The board does not recruit, treat or score. The team does all three. Confidentiality cannot close those routes, and being advisory does not either.

### ASK 14W

Why is the monitoring board shown the arms when the trial team is not?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

the trial team recruits, treats and measures while the trial runs

### VERDICT 75W

Somebody must see unblinded data or a harmful treatment could continue unnoticed. The safeguard is separation of roles. The monitoring board does not recruit participants, deliver treatment or score outcomes. What it learns cannot change those processes directly. The trial team does all three. If that team knows which arm is winning, ordinary judgement can shift enrolment or follow-up. It can also shift how an ambiguous score is written. Confidentiality alone cannot remove those pathways.

### TAKEAWAY 16W

The people who could act on an early result are the people kept from seeing it.

### 4.3 When the blind is opened for one person

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · SEQUENCE

#### SCENE 37W

Diouf has a call from an intensive care registrar at three in the morning, on speaker. A CLARION-3 participant is in front of him and the treatment he is considering depends on what she has been receiving.

#### GUIDE 55W

Four steps, and the rail is not a clock — every one of them could be done in the next ten minutes. Order them by what each one protects: the thing that cannot be recovered if it is skipped goes first, and the thing that can be done afterwards without costing anybody anything goes last.

#### ASK 13W

Order the four steps by what is lost if the step is skipped.

#### BEHIND THE BUTTON 181W

Why the blind exists. It protects the comparison, not the participant. Every person who knows an assignment is a person whose judgement can drift with it — the treating team, the assessor, the participant. The trial's answer depends on that ignorance being intact across four hundred people, and one code being opened costs one of them.

Why the clinical decision comes first anyway. The blind is a research instrument and this is a patient in front of a registrar at three in the morning. If knowing the assignment changes what is safe to give her, the code is opened. The trial gives up one piece of blinding and does not put the participant's immediate care second.

What has to be written down and why the order matters. The reason, the time and who authorised it are recorded before the code is opened rather than reconstructed afterwards. The participant remains in the randomized assigned-arm analysis; the unblinding is an intercurrent event to document, not a reason to delete her. Later analyses can assess whether knowing the assignment could have affected subsequent measurements.

#### TAKES AS READ

each participant's assignment is held in a sealed record that can be opened · blinding, and the several people it can fail for separately — taken as read

#### VERDICT 119W

The first step cannot be recovered, because it cannot be established after the fact. Once the registrar knows the assignment, nobody can show the request rested on clinical need rather than curiosity. The written reason therefore belongs before the code break. Only then is the code opened and the participant treated on what is known. The assignment does not disappear from the randomized analysis: she remains analysed in the group to which she was assigned under the prespecified treatment-strategy approach. What changes is that subsequent measurements may

now have a route for treatment knowledge to affect them, so the unblinding is documented and handled in the estimand and sensitivity analyses. The blind protects the trial; immediate care comes first.

**TAKEAWAY 17W**

The blind exists for the trial; it is opened when a decision for one person needs it.

## 4.4 The plan that was in force — Review

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · CALLBACK · PROTOCOL

### SCENE 41W

Nkechi Umeh, the regulatory lead, has a reviewer's letter asking how CLARION-3 can show its main analysis was fixed before the data existed. Four documents are on her desk, and each one proves something different about when a decision was made.

### GUIDE 52W

Four documents and four things they can prove. Pair them by asking two questions of each: does it carry a date, and could anybody at the sponsor have set that date? A record the sponsor controls is still worth keeping. It simply proves something narrower than a record the sponsor cannot reach.

### ASK 11W

Match each document to what it can actually prove about timing

### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

### TAKES AS READ

a trial's plan can be amended while it is running

### VERDICT 93W

Prespecification is a claim about time, so what supports it is a date somebody outside the sponsor fixed. A registry timestamp is exactly that: the entry went up, and the registry recorded when. A working file proves nothing about timing, because its history belongs to the drive and its contents are whatever they are this morning. Signed minutes prove that a named group agreed something on a stated day, on a record the sponsor keeps itself. An email proves one person's intention. All four are evidence, and they are not the same evidence.

### TAKEAWAY 17W

What a document proves about timing depends on who was able to set the date on it.

**END OF THE DAY 14W**

A trial's small budget for being wrong is spent by looking, not by finding.

## Why is site 12 so fast?

### PLAN CARD 41W

Friday, 17 days out. Site 12 has entered twice its share of patients, and its screening log is far shorter than the rest. Today you work out whether that site is well run, or taking in a different kind of patient.

### BRIEFING 10W

One site's screening log is shorter than every other site's.

### WHAT YOU DECIDE 12W

Work out why site 12 recruits twice as fast as anyone else.

## 5.1 What a short screening log says

SITE OPERATIONS • MONITORS' ROOM • A ROOM • CHOICE

### SCENE 30W

31 screening logs on 1 page. Most sites screen four and enter one. Site 12 screens 11 and enters 9, and its consent rate is the highest in the trial.

### GUIDE 57W

Four explanations for a short screening log. Ask of each what the log would look like if it were true. The high entry rate is not the strongest clue. The nearly empty reasons column is. Eleven screened and nine entered should still record why two were not, and a year should show many who failed before consent.

### ASK 8W

Which explanation best fits site 12's screening log?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

sites record everybody they assess as well as everybody they enter • eligibility criteria, and what widening them changes — taken as read

### VERDICT 77W

The high entry rate is not the strongest clue. The nearly empty reasons column is. A site that truly screened 11 people and entered 9 should still record why the other 2 were not entered. Across a year, it should also show many people who failed before consent. If those people never reach the log, the trial cannot describe who was considered and passed over. The result is a selected group whose recruitment funnel cannot be reconstructed.

### TAKEAWAY 14W

Recruitment figures describe a funnel, and an empty log hides the top of it.

## 5.2 Balance is bought in blocks

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A PERSON · CHOICE

### SCENE 33W

Diouf pulls site 12's kit records. The site has entered 94 participants, 49 to 1 arm and 45 to the other, in a sequence that never runs more than 4 the same way.

### GUIDE 60W

Four purposes for a short block. Ask of each whether it is bought or paid for. Chance is fine over two thousand people and unreliable over ninety. Blocks force the counts to catch up every few entries. Predictability is the price rather than the point, which is why the block length varies and nobody at the site is told it.

### ASK 16W

Why is the sequence built so that no run of the same arm goes past four?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

assignment is by chance, in a sequence prepared before the trial opened

### VERDICT 88W

Pure chance is fine over 2000 people and unreliable over 90. A site that closes at 19 participants could easily have 13 on 1 arm and 6 on the other, and any difference between sites then partly becomes a difference between arms. Working in small blocks forces the counts to catch up every few entries, so the arms are balanced within each site at every point. The cost is predictability, which is why the block length varies and why nobody at the site is told what it is.

### TAKEAWAY 14W

Chance alone is unreliable in small numbers, which is why the sequence is constrained.

### 5.3 Make the three-point offset move

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CONTROL

#### SCENE 38W

Kwame Ansah has matched visit records from site 12 and 3 comparison sites. The symptom score sits about 3 points higher at site 12. A re-scoring console can change one feature of the setup for the same records.

#### GUIDE 66W

The number you are watching is the offset in symptom score between site 12 and the comparison sites, currently about three points. The console can change one feature of how the same records were produced, and the run-to-run noise is about half a point, so anything smaller than that is not a response. Change one, run it, restore it, and name the feature the offset follows.

#### ASK 15W

Which single change makes the offset disappear, and does putting it back restore the offset?

#### BEHIND THE BUTTON 166W

Why the records are held fixed. Nothing about the patients changes in this console: the same visits, the same people, the same dates. Only a feature of how the score was produced changes. That is what makes it a controlled comparison rather than a new study, and it is why a response points at the procedure rather than at the disease.

What the three candidates are for. Who does the scoring, how visit days were matched, and what mix of cases each site saw are three ordinary ways a site can look different without being different. Each is a real explanation somebody at the meeting will offer, and only one of them survives being switched.

Why half a point is the floor. Rescoring the same records twice does not give the same number twice — scales have judgement in them. That scatter is the yardstick every response has to beat, and a difference under it is a measurement of the yardstick rather than of the site.

#### TAKES AS READ

a systematic offset is a repeated shift rather than random scatter · a control test changes one factor while holding the rest fixed · hard endpoints against judged ones, and blinded adjudication — taken as read

#### VERDICT 83W

A site-wide offset can come from the patients, the timing of visits or the person applying the scale. The hard endpoint already argues against a healthier site 12 population. The control test goes further. It keeps the same records and changes one candidate cause at a time. Only the reader change moves the score by far more than the background noise. When the original reader is restored, the offset returns. That reversal is much stronger evidence than a correlation between site and score.

#### TAKEAWAY 20W

A suspected cause becomes persuasive when changing only that factor moves the reading and reversing it brings the effect back.

## 5.4 Eleven a week, through the summer

SITE OPERATIONS • MONITORS' ROOM • A ROOM • HOLD

### SCENE 39W

The trial needs eleven entries a week to finish on time. Sites go on holiday, a competing trial opens at the largest one, and the only lever the office has is where it puts its monitors and its money.

### GUIDE 53W

Hold the entry rate inside the band on the tracker. The band narrows as the summer goes on, because the shortfall that could be made up in September cannot be made up in November. Site support is your control, and every change at a site keeps costing entries until the support answers it.

### ASK 11W

Hold the entry rate at eleven a week through the summer.

### BEHIND THE BUTTON 121W

Why a rate rather than a total. Recruitment is a funnel — screened, eligible, consented, entered — and each stage has its own losses. A site that stops screening does not lose one patient; it loses its share of every week that follows.

Why a shortfall compounds. The trial ends when its events accumulate, and events accumulate from patients already in it. A patient not entered in July is not contributing follow-up in August either, so a week's shortfall costs more than a week.

Why the band tightens. Early in the summer there is time to recover; by the end there is not. The same rate below target is a larger fraction of what remains recoverable with every week that passes.

### TAKES AS READ

a trial enters patients at some rate over time • eligibility criteria, and what widening them changes — taken as read

### VERDICT 155W

Recruitment is a rate and every disturbance to it is a rate too. A site's principal investigator going on leave does not cost one patient; it costs that site's weekly share for as long as the leave lasts. A competing trial at the largest site takes a fraction of every week's screening from then on. A new site opening adds a fraction the same way. None of them is a step, which is why moving a monitor for one week and moving them back changes nothing that lasts. And the shortfall compounds in a way a simple total would not show: an event-driven trial finishes when its events accumulate, and events come from patients already enrolled, so a patient not entered in July is also not contributing follow-up in August. That is why the band narrows through the summer. The same rate below target is recoverable in June and is the trial's timeline in September.

### TAKEAWAY 14W

A shortfall in a rate is not made up by a good week later.

END OF THE DAY 9W

A site cannot see what is unusual about itself.

## The fix that slowed the trial

### PLAN CARD 38W

Monday, 14 days out. The trial widened two entry rules eight months ago to keep patients coming, and the event curve has flattened since. Today you put a number on how much of the trial has actually happened.

### BRIEFING 8W

Recruitment recovered, but the event rate did not.

### WHAT YOU DECIDE 15W

Separate what the amendment demonstrably changed from what the team is only assuming it changed.

## 6.1 How much of the trial has happened

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

### SCENE 34W

Feldman has two numbers on the board: the events CLARION-3 was designed to need, and the events adjudicated so far. Balogun has a third, the months elapsed, and thinks it tells the same story.

### GUIDE 59W

Five numbers, and three of them measure calendar time or people: the months elapsed, the months planned, and the number randomised. Ask of each whether an interim boundary depends on it. A trial three quarters through its years can hold half its events. That is why the meeting moves when the event curve flattens, and the date does not.

### ASK 11W

Estimate the fraction of CLARION-3's information that is in hand today.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

the trial is powered on a number of events rather than a number of people

### VERDICT 93W

Calendar time and information are different quantities, and only one of them decides what an interim can say. A trial three quarters of the way through its planned years may hold half its events, because the people recruited late have been followed for months rather than years and because a healthier group produces events more slowly. The boundary at each interim is set by the information fraction, not by the date, so a flattening event curve moves the meeting without moving the calendar. It is also why finishing takes longer than anyone forecasts.

### TAKEAWAY 17W

The share of the planned events already collected is what says how far through a trial is.

## 6.2 What changed, and what are we assuming?

SITE OPERATIONS · MONITORS' ROOM · A PERSON · BALLPARK

### SCENE 32W

Marchetti has the entry criteria as they were and as they are, with the participants recruited under each. Balogun wants to know whether the amendment cost anything he should have seen coming.

### GUIDE 89W

The amendment brought in people at lower baseline risk, and the observed event-accrual curve flattened. That does not tell us the treatment hazard ratio. The board therefore runs a sensitivity case: if the broadened population also had a smaller relative effect, HR 0.85 instead of the design assumption 0.75, how many events would the same power calculation require? The numerator is the same for both; what changes is the squared logarithm underneath. One tile is that term for the original criteria, which is the comparison rather than the answer.

### ASK 23W

In the sensitivity case where the true hazard ratio is 0.85 rather than 0.75, how many events would the same design formula require?

### BEHIND THE BUTTON 135W

What is actually observed. The broadened population has a lower primary-event rate, so events are accruing more slowly. That can lengthen an event-driven trial even if the relative treatment effect is unchanged.

Why the effect size is squared underneath. The number of events needed goes as one over the square of the log hazard ratio, so a treatment effect that dilutes a little costs a lot. Moving from 0.75 to 0.85 roughly triples the events required, which is the amendment's real price.

What is only a sensitivity assumption. Lower baseline risk does not imply a hazard ratio closer to one. Treatment-effect heterogeneity is a separate question. The HR = 0.85 case below asks what the event requirement would be if the effect were smaller; it is not evidence that the amendment caused that smaller effect.

### TAKES AS READ

entry criteria decide who may be randomised

### VERDICT 121W

For the simplified equal-allocation log-rank planning formula used here, required events scale roughly as  $1/(\ln \text{HR})^2$ . With the original design assumption HR = 0.75, the authored numerator 31.4 gives about 380 events. If, as a sensitivity scenario, the true effect in the broadened population were only HR = 0.85, the same formula gives about 1,190 events. That is a useful stress test because a smaller relative effect is expensive. But it is not an inference from the observed lower event rate. Baseline risk controls how quickly events accrue; the hazard ratio describes relative treatment effect. The amendment can clearly change the first by changing

who is enrolled. Whether it changes the second requires interaction evidence or a prespecified model, not intuition.

**TAKEAWAY 24W**

The amendment demonstrably slowed event accrual; a smaller relative treatment effect is a separate hypothesis to stress-test, not a fact inferred from lower risk.

### 6.3 Changing the plan, on the record

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · SEQUENCE

#### SCENE 30W

Umeh has the amendment file open: the change, the approvals, the registry update and the date each happened. One of the four is a fortnight later than it should be.

#### GUIDE 60W

All four steps will happen, and one of them is a fortnight late. Ask of each what order makes the change defensible. Approved by outsiders, published before use, then handed to sites. That sequence shows the change was not chosen after seeing what it would do. Enrol first and register after, and nothing can ever show the two were independent.

#### ASK 9W

Establish what the amendment obliged the trial to do

#### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

#### TAKES AS READ

the entry criteria are part of the protocol and can be amended

#### VERDICT 82W

The order is the protection. A change made for a stated reason, approved by people outside the trial, published before it is used and only then handed to sites, is a change nobody can have chosen after seeing what it would do to the results. Enrol first and register afterwards and there is no way, ever, to show the two were independent. The audit does not ask whether widening was wise — it asks whether the record was made before it mattered.

#### TAKEAWAY 19W

An amendment is judged by whether it was recorded before it was used, not by whether it was sensible.

## 6.4 Eligible, or nearly

SITE OPERATIONS · MONITORS' ROOM · A ROOM · BELT

### SCENE 34W

The screening log has more entries than the meeting has time for. Some of these people meet the protocol as written. The rest are close, and close is not a category the protocol has.

### GUIDE 40W

Two bins. Somebody is eligible if every criterion in the protocol is satisfied as it is written. Anything that requires reading a criterion generously is not eligible, however sensible the generosity looks. Sort on the protocol, not on the patient.

### ASK 16W

Send each screening entry to the bin that says whether the protocol's eligibility criteria admit them.

### BEHIND THE BUTTON 80W

Why the line is where it is. Every criterion was written to make the trial's population the one the question is about. A patient admitted by a generous reading is in the analysis but not in the population, and nothing downstream can separate them again.

Why "nearly" is the dangerous pile. Nobody enrolls somebody wildly ineligible. Trials are damaged by a hundred small generousities, each defensible on its own, which together make a population nobody planned and no protocol describes.

### TAKES AS READ

a protocol states who may be entered into a trial

### VERDICT 152W

Eligibility criteria are what make a trial's population the population its question is about, and they only work if they are applied as written. The dangerous entries are never the obviously ineligible ones — nobody enrolls a patient twenty years outside the age range. They are the ones a sensible person would let through: an eGFR of 29 against a limit of 30, an ejection fraction measured six weeks ago instead of four, a diagnosis made clinically rather than by the stated test. Each is defensible alone. A hundred of them together produce a trial population nobody planned, whose result applies to nobody in particular, and the analysis cannot separate them out afterwards because the log records that they were eligible. That is why the honest handling of a near-miss is a protocol amendment, prospectively, with the reason written down — and not a decision at the screening desk on a Thursday.

### TAKEAWAY 12W

A criterion read generously is a criterion the trial no longer has.

### END OF THE DAY 22W

Widening eligibility can change event accrual and the target population; it does not by itself prove that the relative treatment effect changed.

## The 91 people without an outcome

### PLAN CARD 38W

Tuesday, 13 days out. 91 patients have no twelve-month outcome, and 48 of them stopped coming after a bad turn was noted in the file. Today you decide what the trial says about the people it has lost.

### BRIEFING 13W

91 people have no outcome recorded, and they are not a random 91.

### WHAT YOU DECIDE 13W

Decide what to do about the participants the trial has lost track of.

## 7.1 Missing is not one thing

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · PROTOCOL

### SCENE 44W

Friday's database lock is now on the wall clock. Four kinds of missing record are still in the queue, each with a count beside it. Tobias Renner, the data management lead, wants one rule for all of them before the next site call starts.

### GUIDE 54W

Four gaps and four consequences. Pair them by asking whether the reason for the gap is connected to how the person was doing. A move abroad is not. Stopping attendance after worsening is. A lost fax batch is recoverable. And one of them closes the door permanently, which is a different problem from either.

### ASK 10W

Match each gap to what it does to the comparison.

### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

### TAKES AS READ

an outcome can be absent for reasons connected to the outcome itself

### VERDICT 86W

Missing data are dangerous for different reasons. A missed visit because somebody moved is mostly a loss of precision. A missing outcome after a participant worsened is different, because the reason for the gap is tied to the thing being measured. That can bend the comparison. A form still sitting at a site is recoverable and should be chased before lock. Withdrawal of consent changes what the trial is allowed to pursue. One rule for every blank cell would hide these differences instead of managing them.

### TAKEAWAY 17W

A gap that depends on what happened is not the same as a gap that does not.

## 7.2 Everyone as assigned

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A PERSON · BALLPARK

### SCENE 29W

Two analyses are on Feldman's desk. One counts everybody in the arm they were assigned. The other counts only those who completed treatment, and it shows a bigger effect.

### GUIDE 62W

Work the relative risk out for the analysis the board is shown — everybody in the arm they were assigned to, whatever they went on to do. Pick the event rate in the treatment arm and the event rate in the control arm and divide. Two of the tiles are the per-protocol rates, which is the other analysis and the other answer.

### ASK 11W

What is the relative risk in the analysis by assigned arm?

### BEHIND THE BUTTON 177W

What relative risk is. The event rate in the treatment arm divided by the event rate in the control arm. Below one means fewer events on treatment. It says nothing about how many people that is — a relative risk of 0.7 can be six fewer events per hundred or six per hundred thousand, which is why it needs the absolute difference beside it.

Why randomized groups remain the starting point. Randomisation protects baseline comparability in expectation, including for factors nobody measured. Selecting participants after assignment on treatment completion or follow-up can reintroduce post-randomization differences. That does not make every as-assigned analysis automatically correct: the analysis still has to define how discontinuation, rescue treatment and missing outcomes enter the treatment effect being estimated.

Why the two analyses answer different questions. The assigned-arm treatment-policy analysis estimates the effect of assignment to the treatment strategy under the trial conditions, including the consequences of discontinuation as specified in the estimand. A completer-only comparison asks about a selected post-randomization subset and can be biased if completion depends on treatment or prognosis.

### TAKES AS READ

some participants do not take the treatment they were assigned

### VERDICT 115W

Relative risk is  $RR = p_{\text{treatment}} / p_{\text{control}}$ . In the assigned-arm analysis, 84 of 600 on treatment had the event and 120 of 600 on control did, giving  $0.140/0.200 = 0.70$ . Among completers the authored rates give about 0.60, but that subset was defined after randomization and completion is related to what happened during treatment. The larger apparent benefit therefore cannot simply replace the randomized comparison. The assigned-arm analysis preserves the original groups, while the protocol must still specify how intercurrent events and missing outcomes are handled. Neither relative risk says the absolute scale: here the raw 12-month risk

difference is six percentage points, corresponding to an NNT of about 17 at that stated horizon.

**TAKEAWAY 28W**

Keeping people in their randomized groups preserves the baseline comparison; post-randomization events and missing outcomes still have to be handled in a way that matches the trial's estimand.

### 7.3 Forty hours before the lock

SITE OPERATIONS · MONITORS' ROOM · A ROOM · ALLOCATE

#### SCENE 38W

Lucia Marchetti has a fortnight, 31 sites and 40 monitoring hours she can still deploy. Four recoverable gaps compete for that time. 14 people who withdrew consent are locked out of the plan and cannot be contacted again.

#### GUIDE 59W

Forty monitoring hours, four recoverable gaps, and a fortnight before the lock.

Watch the questions list rather than the site map: it shows what the interim analysis can still answer under your plan. The fourteen who withdrew consent are locked out and cannot be contacted, so no plan recovers them — spend the hours on what is still recoverable.

#### ASK 11W

Which monitoring jobs buy the answers the interim analysis actually needs?

#### BEHIND THE BUTTON 147W

What makes a gap recoverable. A missing form that exists on paper, a visit whose data was never entered, or a query nobody answered can often be closed by somebody going to look. A participant who has withdrawn and asked for no further contact cannot be chased for new information. Data already collected are handled according to the consent, protocol and applicable rules; withdrawal does not automatically erase them.

Why the interim analysis defines value. Not every gap affects it. A missing secondary endpoint at a small site may not move anything, while a handful of missing primary events at a large one can decide whether the trial continues.

Why hours rather than sites. Monitoring time is the resource, and sites differ enormously in how long a gap takes to close. Four hours at a well-run site can recover more than a day at one in difficulty.

#### TAKES AS READ

missing outcomes are most dangerous when the reason for missingness is tied to the outcome · withdrawal of consent prevents further contact

#### VERDICT 79W

The dangerous gaps are not simply the largest ones. 48 participants disappeared after worsening, so their missingness is tied to the outcome itself. That can bend the comparison if the losses differ by arm. Seven outcomes already exist on paper and are cheap to restore. People who moved are missing for a more neutral reason, so tracing them mainly improves precision. Rechecking complete records buys less. The allocation should protect validity first, then spend any remaining effort on precision.

#### TAKEAWAY 17W

A monitoring plan is a set of questions it can answer and questions it deliberately leaves unanswered.

#### END OF THE DAY 19W

What you do about missing outcomes is a decision, and it has to be made before you see them.

## Forty-one hours too warm

### BEFORE THE DAY — HUNT

#### Six unblinding envelopes, and the safe holds four

Every unblinding envelope has to be accounted for before the committee meets, and two are not in the safe. An envelope somebody could have opened is a broken blind whether or not anybody opened it, and the trial is defended on the record rather than on goodwill.

### PLAN CARD 42W

Wednesday, 12 days out. The fridge logger at site 19 shows 41 hours above the safe range in July, and the alarm was signed off twice. Today you decide what a warm fridge means for the nine patients who took that kit.

### BRIEFING 13W

A freezer alarm at one site was silenced for two days in July.

### WHAT YOU DECIDE 14W

Work out which kit is compromised and what that obliges the trial to do.

## 8.1 Kit that cannot be vouched for

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A ROOM · CHOICE

### SCENE 36W

Site 19 has already dispensed 11 boxes from the warm fridge to 9 participants. Farida Diouf has the logger trace, dispensing list and stability sheets open. A replacement shipment will not reach the site until morning.

### GUIDE 57W

Four readings of a warm fridge. Ask of each whether the fault can act on both arms equally. Identical boxes can hold different contents, and heat has nothing to damage in a comparator. So the direction of the error is not symmetric. Eleven boxes is a question about size, which is separate from the question about direction.

### ASK 15W

What is the most important effect of the site 19 excursion on the trial comparison?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

both arms receive kit that looks identical

### VERDICT 79W

Identical boxes can hide different contents. Heat can damage active drug, while the comparator has no active material to lose. The excursion can therefore pull treated participants toward the control result. That is different from an error that affects both arms equally. 11 boxes are unlikely to change the whole trial, but the scale of the problem is a separate question from its direction. Recording the event lets later reviewers distinguish a small one-sided fault from a repeated one.

### TAKEAWAY 12W

A storage failure damages one arm's material and leaves the other untouched.

## 8.2 What the nine are owed

SITE OPERATIONS · MONITORS' ROOM · A PERSON · TRIAGE

### SCENE 29W

Nine people received kit from that fridge. Marchetti has to tell site 19 what to say to them, and Doyle has asked to see the wording before it goes.

### GUIDE 65W

Four things to tell nine people, and two duties pulling opposite ways. Ask of each whether it can be said to all nine equally without saying what is in the box. Saying they may have been under-treated says the drug is theirs. Saying nothing because the arms are unknown puts the blind above the person, which is the inversion the whole trial exists to avoid.

### ASK 9W

What should the nine participants be told this week?

### BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

### TAKES AS READ

participants are told about problems that affect their own treatment · allocation  
concealment: protecting the moment of entry — taken as read

### VERDICT 92W

Two duties pull in different directions and both can be met. People are entitled to know about a fault that touched their own treatment, and telling them costs the trial nothing if the wording covers all nine equally. Saying they may have been under-treated says the drug is in the box. Saying nothing because the arms are unknown treats the blind as more important than the person, which is the inversion the whole trial is arranged to avoid. Waiting for the board is 12 more days for information that is already certain.

### TAKEAWAY 18W

A participant is owed what affects them, said in a way that does not tell them their arm.

### 8.3 What the excursion record can actually prove

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · ATTEST

#### SCENE 36W

Nkechi Umeh has the site 19 excursion form, the logger trace, dispensing records and an inventory statement. Four claims on the form are signed. She has time to verify only two before the deviation file closes.

#### GUIDE 59W

Four signed claims on the excursion form, and time to verify two. Open each and read what backs it: the logger trace, the dispensing records, the inventory statement, or the form itself. Hold what the evidence does not support, and verify where a wrong claim would most damage somebody's later judgement about whether the drug was fit to use.

#### ASK 14W

Which claims most need verification before this record can support somebody else's judgement later?

#### BEHIND THE BUTTON 146W

What an excursion record is for. It documents a period when storage went outside its specified range, so that somebody later can decide whether the product was affected. It is written to be read by a person who was not there, which is exactly why the backing matters more than the wording.

Why the logger trace is not automatically the strongest evidence. It records what the logger saw at its position. Whether that position represents the product, whether the logger was calibrated, and whether the affected stock was in that location at that time are separate claims — and they are the ones usually signed rather than checked.

Why two checks and not four. The deviation file closes, and after that the record is what a reviewer sees. A claim held as unverified is honest; a claim verified after the file closes is a second deviation.

#### TAKES AS READ

a deviation record is evidence for later review · a signed statement records what someone asserts, not necessarily what the underlying records show

#### VERDICT 84W

A deviation form does not become reliable because somebody signed it. The duration can be checked against the logger, and the nine recipients can be checked against dispensing records. The important weak link is scope. If more than 11 kits were in the fridge, the size of the problem changes immediately. That claim needs an inventory reconciliation, not another signature. Recording the excursion still matters even if the final effect is small, because later reviewers can only challenge a judgement when its evidence survives.

#### TAKEAWAY 18W

A signed record is useful only when the claims that could change the conclusion are tied to evidence.

## 8.4 A trial ends at a number of events — Review

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · CALLBACK · BALLPARK

### SCENE 45W

Dr. Miriam Feldman, the trial statistician, has been asked to sanity-check a schedule for HELIOS, the smaller trial starting in this building. Its sponsor has written a finish date on the front page. Feldman puts three of the trial's own numbers on the whiteboard instead.

### GUIDE 57W

Five numbers, and three of them belong to other questions. Ask of each whether this schedule depends on it. The events adjudicated so far say how far HELIOS has got rather than how long it has left. The count of hospitals sets how fast people are recruited, which is a different arithmetic from how fast events arrive.

### ASK 22W

HELIOS is an event-driven trial too. Estimate how long it has to follow its participants to reach the planned events it needs.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

the trial compares how often an event happens in two arms · events accumulate over time, at some rate per participant per year

### VERDICT 96W

A trial that waits for events finishes when the events arrive. Nine hundred participants who stay well carry almost no information about which arm is better; the comparison rests on the ones something happened to. So the design fixes a number of events, and the calendar follows from the rate those events arrive at, which belongs to the participants rather than to the sponsor. A finish date written before the rate is known is a forecast in the clothes of a commitment, and it is why progress is reported as events accumulated rather than months elapsed.

### TAKEAWAY 20W

A finish date is an output of the event rate, and the event rate is not the sponsor's to set.

### END OF THE DAY 17W

A material problem becomes an analysis problem the moment it is not spread evenly across the arms.

## The blind has leaked

### PLAN CARD 41W

Thursday, 11 days out. Site doctors were asked to guess which arm each patient was on, and across 1,180 guesses they were right 71% of the time. Today you work out what a guessing rate like that does to each outcome.

### BRIEFING 16W

Investigators were asked to guess the arm, and they were right far more often than chance.

### WHAT YOU DECIDE 14W

Find out whether the people scoring the outcome knew which arm they were scoring.

## 9.1 The blind that had already gone

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CHOICE

### SCENE 34W

At the annual visit, investigators and participants guessed each person's arm. Investigators were correct 71% across 1,180 guesses. The case packet also contains first-fortnight symptom logs, the allocation audit and the adjudicated outcome summary.

### GUIDE 56W

Four explanations for guessing right 71 per cent of the time. Ask of each whether it accounts for both investigators and participants guessing well. A leaked list would not. A treatment effect big enough to be visible in improvement is not in the event data. And chance across 1,180 guesses does not reach anywhere near 71.

### ASK 13W

What best explains investigators guessing the treatment arm correctly 71% of the time?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

blinding is meant to keep the person measuring from knowing the arm

### VERDICT 74W

71% correct across 1,180 guesses is far from the 50% expected by chance. The visible flush also supplies a direct mechanism. Investigators can see it before they later score symptoms. A leaked allocation list could also raise guessing, but it would not explain participants guessing well too. Guessing from improvement would require a treatment effect much larger than the event data show. The survey therefore measures a real loss of blinding, not random noise.

### TAKEAWAY 15W

When a treatment announces itself, the people measuring stop being blind whatever the protocol says.

## 9.2 Which measurements share the leak

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · TRACE

### SCENE 36W

Feldman has 4 CLARION-3 measurements on the wall. The guessing survey says the blind has leaked. Each measurement came through a different path, and the sponsor wants to know which evidence still stands on its own.

### GUIDE 52W

Open each of the four measurements to see what it came through. Keep the ones that would still stand if the blind has leaked, and untick the rest. Then name the source the compromised ones share. A leak does not spoil everything equally, and the sponsor needs to know exactly what survives.

### ASK 13W

Which measurements remain independent of the leak, and what source compromises the others?

### BEHIND THE BUTTON 132W

What a blind is protecting. If people know which arm a patient is in, that knowledge can reach the measurement: through how symptoms are elicited, how ambiguous events are coded, how closely anybody is followed up. Blinding is what stops the treatment assignment influencing the record of what happened.

Why the guessing survey matters. Asking staff to guess the assignment measures whether the blind held. Guesses much better than chance mean it did not, which puts every measurement that passed through a person who knew under suspicion — and leaves the automated ones alone.

Why withdrawing everything is the wrong response. It discards measurements the leak could not have touched, and it tells a data monitoring committee that nothing is known. Naming what is compromised is what lets the trial continue honestly.

### TAKES AS READ

the visible flush can reveal treatment assignment · adjudication removes the arm and site before the committee rules

### VERDICT 82W

A blinding failure does not contaminate every number automatically. Bias needs a path. The treating clinician can see the flush, infer the arm and then write the symptom score. Participants can see the same flush and change how they take or return kit. The adjudication committee sees neither arm nor site, so that route is cut. The national death register is farther away still. Dependency tracing shows why four results pointing the same way are not automatically four independent pieces of evidence.

### TAKEAWAY 19W

Agreement counts as independent evidence only when the measurements do not share the same route for bias to enter.

### 9.3 Telling the board what it does not yet know

SAFETY MONITORING · MONITORING BOARD ROOM · A PERSON · PROTOCOL

#### SCENE 36W

Vogt reads the interim pack in 11 days, and the PDF is queued for sign-off tonight. The guessing survey is absent because nobody expected it to matter. Balogun wants to leave it for the annual report.

#### GUIDE 64W

Four things about the trial's condition, and four ways a pack can carry them. Match each to the treatment that lets the board weigh it — some belong in the results section, one belongs in a limitations paragraph the board reads before the numbers, and one is genuinely an annual-report item. What decides the match is whether the board's next decision turns on it.

#### ASK 15W

Match each fact about the trial's condition to the way tonight's pack should carry it.

#### BEHIND THE BUTTON 146W

What the board is for. It is the only body that sees the arms separated, and it decides whether the trial continues. Everything it decides rests on the pack in front of it, so a weakness the pack does not mention is a weakness that cannot enter the decision.

Why a limitation is not an admission. A blinding survey suggesting the assignment may be guessable changes how much weight the effect estimate carries. Reported beside the estimate, it is information. Held for the annual report, it becomes something the board was not told before it decided, which is a different document to have to write.

What genuinely can wait. Enrolment running two months behind is a management problem with no bearing on whether the accumulated data support continuing. The test is not importance in general — it is whether this board's next decision turns on it.

#### TAKES AS READ

the board decides whether the trial continues

#### VERDICT 182W

A monitoring board's decision is made out of the pack, so the placement of a fact is not presentation — it decides whether the fact enters the decision at all. The blinding survey qualifies every efficacy number in the document, which is why it goes before them rather than after: a board that reads an estimate and then learns the blind may have leaked has already formed a view. Deviation rates belong site by site in the body, because a comparison is only as good as the sites feeding it and an average hides the two that are the problem. The event count is not a footnote either — 246 of 380 is the information fraction, and it fixes how much any estimate in the pack can settle. Recruitment running behind is real and belongs in the annexe, because nothing the board decides tonight turns on it. What makes the survey the

hard case is that leaving it out is defensible one item at a time and indefensible as a decision: the board cannot weigh a weakness it has not been shown.

**TAKEAWAY 13W**

A board cannot weigh a result whose weakness it has not been shown.

**END OF THE DAY 13W**

A blind that has already failed cannot be restored by insisting on it.

## Does the benefit survive the leak?

### PLAN CARD 38W

Friday, ten days out. The committee has re-read every unclear file with the arm stripped out, and the effect there is smaller than on the symptom score. Today you say what is left, and how tightly it holds.

### BRIEFING 10W

Take the compromised score out and see what is left.

### WHAT YOU DECIDE 11W

Re-read the trial on the outcome the unblinding could not reach.

## 10.1 Reading the same files twice

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CLOUD

### SCENE 30W

The committee has re-read 400 files. Ansah is asked whether a second reading by the same three people, who now know a guessing survey exists, can be trusted at all.

### GUIDE 63W

The cloud is the 400 files, each plotted as the difference between its first and second reading. Place the middle bar and the two sigma bars to report where the differences centre and how widely they scatter, then buy what the committee can actually do to tighten the agreement. The band drawn across the plot is how much drift the endpoint definition tolerates.

### ASK 20W

Report where the first-to-second-reading differences centre and how far they scatter, and get the spread inside what the definition tolerates.

### BEHIND THE BUTTON 153W

What a difference of zero would mean. That the rule decided the reading, not the reader. A centre away from zero is drift — the second pass systematically scoring differently from the first — and a wide scatter is inconsistency, which is a different defect with a different fix.

Why the two are not the same problem. Re-centring a drifted committee is a recalibration: restate the rule, define the ambiguous case, and the middle moves. Only a sharper rule narrows the spread, and no amount of re-centring does it, which is what this panel is really about.

Why knowing the survey exists is dangerous. Three people who now expect the blind may have leaked are three people with a reason to read the ambiguous files differently, and their agreement with each other would not show it. Consistency between two readings comes from the rule, not from memory of what they decided last time.

### TAKES AS READ

the adjudication committee never sees the arm · hard endpoints against judged ones, and blinded adjudication — taken as read

### VERDICT 164W

A re-read is worth something only if the rule did the deciding. Plotted as first reading minus second, 400 files scatter around a centre near 0.6 events per hundred, with a spread of about 2.9. That is a committee both slightly drifted and quite inconsistent. Those are two defects, and they have different repairs. Restating the definition moves the centre and leaves the width alone. Defining the one ambiguous case narrows it. A fourth reader who has never seen these files narrows it again. Apply every re-centring and no rule change, and the spread sits exactly where it was. That is the lesson. Agreement between two readings by the same people can be manufactured by memory. Only a rule that decides the ambiguous case the same way

twice makes it evidence. The guessing survey matters for that reason. Three readers who suspect the blind has leaked have a reason to read the ambiguous files differently, and agreeing with each other would never reveal it.

**TAKEAWAY 11W**

Consistency between two readings comes from the rule, not from memory.

## 10.2 How big, and how wide

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

### SCENE 45W

The press office has lifted the prespecified 12-month risk difference, “4.2 percentage points,” into a draft briefing. Feldman circles the interval beside it. Before the number leaves the room, she wants the team to put both clinical size and sampling noise on this fixed-horizon risk.

### GUIDE 60W

Five numbers, and two of them are answers to other questions: the risk ratio, and the risk difference already worked out. Ask of each what this calculation needs. The number needed to treat is built from the absolute difference, not the ratio. And precision is a separate question from size, which is why the interval is circled beside the headline.

### ASK 19W

At the prespecified 12-month horizon, estimate the NNT and the simple binomial standard error of the 18.9% control-arm risk.

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

the 2,400 participants are allocated about equally between 2 arms · a proportion estimated from a sample has sampling error · effect measures: relative risk, absolute risk reduction, number needed to treat — taken as read

### VERDICT 81W

The risk difference and NNT describe the size of the observed benefit. Precision is a different question. For a proportion, standard error depends on both the event rate and the number measured. With about 1,200 people in the control arm, an 18.9% rate has an SE near 1.1 percentage points. A 95% margin is about twice that. Increasing the sample narrows this uncertainty only with the square root of  $n$ , so halving the SE takes about four times as many people.

### TAKEAWAY 20W

Effect size says how large the difference is; standard error says how much the estimate would move across repeated samples.

### 10.3 What to do about a blind that leaks

RANDOMISATION & BLINDING · KIT WAREHOUSE & COLD ROOM · A PERSON · CHOICE

#### SCENE 30W

600 participants are still in follow-up and the flush has not stopped happening. Diouf has been asked whether there is anything the pharmacy can do about it at this stage.

#### GUIDE 65W

Four options for a blind that has already leaked. Ask of each what it could still protect. Nothing recovers the blind for anybody who saw the first fortnight. What can be protected is the route from belief into the number. Matching the comparator is the right design, four years too late. And a limitation paragraph fixes nothing while six hundred people are still being scored.

#### ASK 10W

What is the best available fix for the remaining follow-up?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

#### TAKES AS READ

the drug produces an early visible effect that the comparator does not

#### VERDICT 95W

The blind has already failed for anybody who saw the first fortnight, and nothing recovers it for them. What can still be protected is the route from belief into the number: separate the assessor from the treating team, and the score is written by somebody with nothing to infer from. Matching the comparator's appearance is the right design and it is four years too late to introduce here. Leaving it in the notes matters — the notes are the source record — and a limitation paragraph fixes nothing while 600 people are still being scored.

**TAKEAWAY 15W**

When a treatment cannot be hidden, protect the measurement instead of pretending the blind holds.

**END OF THE DAY 13W**

A finding that shrinks when the bias is removed was partly the bias.

## Freeze the evidence

### PLAN CARD 34W

Monday, six days out, and nothing is wrong. Open queries are down to 41 and the file is locked on Thursday. Today you close the file, and say what a locked database really promises.

### BRIEFING 9W

No crisis today, only a database lock on Thursday.

### WHAT YOU DECIDE 8W

Close the file that the board will read.

## 11.1 Freezing something that was moving

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · CHOICE

### SCENE 37W

The database-lock checklist is projected on the wall and the cut is three days away. Umeh has one question from the sponsor and another from Renner: once the file freezes, what exactly would justify opening it again?

### GUIDE 63W

Four things a lock might buy. Ask of each whether a frozen file could deliver it. It cannot make the data correct, and it cannot stop collection. What it does is put a visible boundary around what the analysis used, so anybody can rerun it. Reopening is possible and creates a new version of the evidence, which is why it needs a reason.

### ASK 9W

What does locking the database buy the interim analysis?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

data arrive from sites continuously while a trial runs

### VERDICT 90W

A database lock creates a version of the trial that cannot quietly move under the analysis. Queries are closed, coding is fixed and the dataset is frozen before the planned calculation runs. That makes the result reproducible later. Reopening is possible, but it must be exceptional and documented because it creates a new version of the evidence. A discovered error that changes a primary event can justify that step. A cosmetic correction usually cannot. The point is not permanence; it is a visible boundary around what the analysis actually used.

### TAKEAWAY 15W

Locking fixes what the analysis was run on, so the result can be checked later.

## 11.2 Three days, forty-one queries

SITE OPERATIONS • MONITORS' ROOM • A PERSON • DELEGATE

### SCENE 39W

41 queries remain and the interim file locks in 3 working days. One concerns a primary-event date, one a duplicated visit, and 39 are contact-field corrections. Renner has two people available, and you can keep only one problem yourself.

### GUIDE 49W

Forty-one queries, three working days, two people, and one problem you keep yourself. For each thing you hand over, name the owner, the first action, and what brings it back to you. Thirty-nine of the queries are contact-field corrections, which is a volume problem. The other two are not.

### ASK 17W

What do you keep, what do you hand off, and what must each delegated owner bring back?

### BEHIND THE BUTTON 133W

Why volume and judgement are different work. Thirty-nine identical corrections need somebody methodical with a clear rule for the ambiguous ones. A primary-event date and a duplicated visit each need somebody who understands what the field means, because getting them wrong changes the analysis rather than the tidiness of the file.

Which to keep. The primary-event date is the one that feeds the interim analysis directly. A wrong date can move an event across the analysis boundary, which is the one error in this list that changes a result rather than a record.

What a return condition looks like here. Not 'come back if stuck'. A rule: any correction that would change a date, or any duplicate that cannot be resolved from the source document, comes back to you rather than being decided.

### TAKES AS READ

an event outside the cut is absent from this interim analysis • delegation needs an owner, a first action and a condition for returning the problem

### VERDICT 85W

The biggest pile is not the most urgent problem. A missing event date can decide whether a primary event exists in Thursday's analysis at all. Once the file is cut, that omission cannot be repaired for this interim. The duplicate visit is important but stable, and a site monitor can reconcile it against the source note. Contact-field corrections are numerous and visible, yet they do not change the endpoint count. Good delegation protects the irreversible decision while giving stable work clear owners and return rules.

### TAKEAWAY 18W

Urgency belongs to the problem whose consequence becomes irreversible first, not to the problem making the most noise.

## 11.3 Nothing left in the tray

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CHOICE

### SCENE 32W

Ansah has 19 files still under adjudication and 3 days. 2 of the 19 are from participants recruited in the first year and have been waiting on 1 missing discharge summary each.

### GUIDE 59W

Four reasons the tray has to be empty. Ask of each what an unadjudicated file actually is in the analysis. Adjudication is what turns something that happened into a countable event, so nineteen files in a tray are nineteen events nobody had. And files wait longest where the evidence is thinnest, which is not evenly spread across the arms.

### ASK 13W

Why must the adjudication committee clear its tray before the file is locked?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

an event is only counted once the committee has ruled on it

### VERDICT 85W

Adjudication is not paperwork after the fact; it is the step that turns something that happened into a countable event. 19 files sitting in a tray are 19 events the analysis will behave as though nobody had. That is not neutral: files wait longest where the evidence is thinnest, and the thin files are not evenly spread between arms. Rulings made after the lock go into the next version, which means the interim was read on a trial that was quietly missing part of itself.

### TAKEAWAY 12W

An unadjudicated event is invisible to the analysis, whatever the notes say.

END OF THE DAY 12W

A locked file is a promise about a moment, not about perfection.

## Three alarms before nine

### PLAN CARD 43W

Tuesday, five days out, and three things landed before nine. A subgroup crossed its line, a Lyon team published a rival trial, and the regulator wants the liver cases by Friday. Today you decide what goes in the pack, and in what order.

### BRIEFING 15W

A subgroup has crossed its line, a rival has published, and the regulator has written.

### WHAT YOU DECIDE 13W

Rank what the board is told when everything arrives on the same morning.

## 12.1 The subgroup that crossed

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

### SCENE 32W

Participants over 70 show a difference that clears the line on its own. Feldman asks which of 14 prespecified subgroups this is, and whether anybody has looked at the other 13 yet.

### GUIDE 57W

Put an interval around the over-70 result before deciding what it is worth. The interval is the estimate plus and minus about two standard errors, so pick the multiplier and the standard error and work out the half-width. One tile is the standard error of the whole trial, which is the number this subgroup does not have.

### ASK 14W

How wide is the interval around the over-70 result, either side of the estimate?

### BEHIND THE BUTTON 167W

What the width says. A confidence interval is the estimate give or take about two standard errors, and the standard error grows as the group shrinks. A subgroup of a few hundred therefore carries an interval several times wider than the trial's, which is why subgroup estimates swing so far from the overall result.

Why 14 subgroups is the other half of the problem. Testing fourteen slices at the usual threshold gives about a one-in-two chance that at least one clears the line by luck alone. A subgroup that crossed is therefore what an honest trial expects to see, not evidence that something is different about people over 70.

What the over-70 result is for. Its nominal interval is just persuasive enough to tempt a treatment claim. The discipline is to remember that it is one row in a family of subgroup analyses, measured on the outcome most exposed to the blinding failure. That makes it a hypothesis to test, not a treatment recommendation for older participants.

### TAKES AS READ

a subgroup is a slice of the participants analysed on its own

### VERDICT 122W

A nominal 95% interval is estimate  $\pm 1.96 \times \text{SE}$ . With an absolute risk reduction of 6.0 percentage points and SE 2.7, the half-width is about 5.3 points, so this single subgroup's nominal interval is roughly 0.7 to 11.3 and just excludes zero. That is exactly why the result is tempting. But the interval is for one test considered alone. Fourteen prespecified subgroup looks create a multiplicity problem, and the subgroups are correlated because they overlap. The proper question is therefore not whether this one row has  $p < 0.05$ , but whether the analysis plan controlled the family of subgroup claims and whether there is credible evidence of treatment-effect heterogeneity. The finding is hypothesis-generating unless that stronger standard was prespecified and met.

### TAKEAWAY 18W

A subgroup result is one of many results, and it has to be judged as one of many.

## 12.2 The update that is due Friday

SAFETY MONITORING · MONITORING BOARD ROOM · A PERSON · CHOICE

### SCENE 29W

The regulator wants the liver cases written up by Friday. There are now four, not two. Vogt has seen all four by arm; nobody else in the building has.

### GUIDE 62W

Four versions of Friday's update. Ask of each whether a reader could work out how common the problem is. Four cases among forty exposed people is a different thing from four among twelve hundred. So a numerator without a denominator is not a rate. Causality opinions can go in and do not replace the counts, and efficacy belongs to a closed review.

### ASK 9W

What must Friday's update contain to be worth reading?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

serious events are reported to regulators on a stated schedule

### VERDICT 79W

Four cases do not say how common the problem is. 4 cases among 40 exposed people mean something very different from 4 among 1200. The regulator needs both the numerator and denominator, plus enough clinical detail to recognise a pattern. Causality opinions can be included, but they do not replace the counts. Efficacy results belong to the board's closed review. A recommendation about continuing also comes later, after the people responsible for that decision have the same safety evidence.

### TAKEAWAY 14W

A safety update reports the events and the denominator, not a conclusion about them.

## 12.3 What the Lyon paper is worth tonight

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · VALUE

### SCENE 36W

The Lyon trial published a large benefit at 6:10 this morning. By lunch, Balogun wants CLARION-3 to respond. Nkechi Umeh has the paper, its supplement and four review hours before she must state the trial's position.

### GUIDE 51W

Four review hours before a position has to be stated, and a paper published this morning. Open each piece of the Lyon evidence and ask what would change about CLARION-3's position depending on how it comes out. Buy what could change the position, not what would explain the headline more thoroughly.

### ASK 14W

Which pieces of the Lyon evidence are worth buying before tonight's position is written?

### BEHIND THE BUTTON 132W

Why the headline is rarely the useful part. A large benefit in another trial matters to this one only through specifics: the population, the endpoint definition, the comparator, whether the effect is plausible in the patients CLARION-3 enrolled. Those live in the supplement rather than the abstract.

What a position actually has to survive. Balogun will be asked whether CLARION-3 should continue, and the answer has to hold whether the Lyon result replicates or not. Evidence that would only be reassuring if Lyon is right is not worth an hour.

Why four hours is the whole point. There is time to read some of it properly or all of it badly. Choosing which parts is the same judgement every VALUE stop asks for, on the day it is least comfortable to make.

### TAKES AS READ

a related trial can change the uncertainty that makes continued randomisation defensible · an outside result does not rewrite a prespecified analysis plan

### VERDICT 82W

A striking outside result is neither ignored nor copied into the current trial. First ask whether it measured the same kind of outcome, on enough events, with uncertainty narrow enough to matter. Then ask whether its randomisation protected the comparison. Those features determine how much the paper should change the board's uncertainty. A plausible mechanism or an impressive subgroup can support interpretation, but they cannot rescue an incomparable endpoint. The result belongs in the board's discussion, while CLARION-3's registered analysis remains fixed.

### TAKEAWAY 19W

External evidence matters in proportion to how much it can change the decision, not how dramatic its headline sounds.

## 12.4 What needs extra escalation today

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · SPOT

### SCENE 50W

Every serious adverse event is already in the sponsor's standard safety system. What changes on this desk is the extra same-day alert rule for the liver signal. It starts with serious unexpected cases. Then it widens to anything matching the cluster. Then to anything the regulator has asked to see.

### GUIDE 48W

Escalate what the current focused-signal rule requires. The underlying SAE reports are already in the safety system; this board is deciding which cases need an additional same-day alert or focused follow-up. The rule changes as the signal changes, so the scored moments are either side of each update.

### ASK 15W

Escalate to the focused safety rule in force, while ordinary SAE reporting continues underneath it.

### BEHIND THE BUTTON 104W

Why the escalation rule moves. Standard safety reporting continues underneath this game. The additional signal-management rule broadens when a cluster appears: after the second similar liver case, even non-serious matching cases may deserve rapid aggregate review; after a regulator asks for a defined phenotype, that request sets the focused follow-up timetable.

Why the distinction matters. A case not selected for this extra alert is not being hidden or left unreported. It remains in the ordinary safety system. The game is about recognizing when new information changes the threshold for additional escalation, not about inventing a rule that some adverse events can simply be ignored.

### TAKES AS READ

a trial reports certain events without waiting to decide what caused them

### VERDICT 107W

Three focused-escalation rules run across the day. The first alerts on serious unexpected cases. The second widens to anything resembling the emerging cluster. The third covers anything the regulator has asked about. One report can satisfy two rules at once, which is what makes each change cost something real. The panel scores the window either side of each change. Most of the queue is wanted by neither rule and is correctly held for ordinary assessment. The hour after a second case appears is the hard one. The rule has already widened, and the reports still going out are the ones that were serious-and-unexpected when the morning started.

### TAKEAWAY 17W

Standard reporting continues underneath the crisis; the changing rule only determines which cases receive extra same-day escalation.

Three urgent things are not one big thing, and the board is owed the difference.

## The result that looks too good

### BEFORE THE DAY — CANVASS

#### Whether the sites are actually using the new consent form

The amendment went out three weeks ago and the analysis assumes it is in use. Ask round the sites until enough answers have come back to say whether they switched, because a trial half-consented under two versions has a problem nobody can fix retrospectively.

### PLAN CARD 35W

Wednesday, four days out. The plan named 14 subgroups before the trial began, and the over-70 group is number eleven. Today you work out how often 14 looks throw up one false positive by luck.

### BRIEFING 14W

The list of subgroups the plan named in advance has 14 entries on it.

### WHAT YOU DECIDE 14W

Work out what the over-seventies result would have cost if it had been believed.

## 13.1 How often something clears the line by luck

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A ROOM · BALLPARK

### SCENE 34W

14 prespecified subgroups, each tested at the usual 5%. Feldman writes the arithmetic on the board and asks what the chance is that at least one clears the line with no real difference anywhere.

### GUIDE 56W

Five numbers, and two of them belong elsewhere: the number randomised, and the one that stands for certainty. Ask of each whether this stacking depends on it. Each test carries its own small chance of misfiring, and those chances multiply rather than staying small. Real subgroups overlap, which softens the number without changing what it says.

### ASK 28W

Assume for a moment that the 14 subgroup tests were independent. What is the benchmark chance that at least one looks statistically significant at 5% by luck alone?

### BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

### TAKES AS READ

each test has a 5% chance of a false positive on its own

### VERDICT 82W

As an independence benchmark, fourteen separate 5% tests give  $1 - 0.95^{14} \approx 0.51$  for at least one false positive. Real subgroup tests are usually correlated because the same participants appear in several slices, so 51% is not CLARION-3's exact familywise error rate. The exact rate depends on those correlations and on the multiplicity method in the analysis plan. The point is structural: a nominal p-value from one selected row cannot be interpreted as though that row were the only question asked.

### TAKEAWAY 15W

The chance of at least one false positive rises fast with the number of tests.

## 13.2 The slice, and what it was scored on

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A PERSON · CHOICE

### SCENE 33W

The over-seventies result is on the symptom score rather than the adjudicated event. Ansah points out that this is the outcome the guessing survey landed on, and that older participants attend most often.

### GUIDE 57W

Four accounts of what the subgroup result rests on. Ask of each whether the outcome named is the protected one or the leaky one. Two weaknesses compound here: one of fourteen looks, measured on the score the guessing survey reached. Older participants attend more often, so there are more scored visits for a small bias to enter.

### ASK 7W

What does the over-seventies result rest on?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

the symptom score is written by the treating doctor and the event is adjudicated

### VERDICT 80W

Two weaknesses compound here. The finding is 1 of 14 subgroup looks, so chance already has many opportunities to produce something interesting. It is also measured on the symptom score, which the guessing survey showed can inherit treatment knowledge. Older participants attend more often, creating more scored visits where a small bias can enter. The adjudicated event in the same subgroup shows nothing unusual. That contrast matters: the subgroup signal is strongest exactly where the measurement process is least protected.

### TAKEAWAY 13W

A subgroup finding inherits every weakness of the outcome it was measured with.

### 13.3 What to say about a slice

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · CHOICE

#### SCENE 30W

Vogt will see the over-seventies table whatever the pack says about it. What the pack can do is tell her what she is looking at before she looks at it.

#### GUIDE 59W

Four ways to present one slice, and the board will see the table whatever the pack says. Ask of each what reaction the layout invites. A striking slice alone invites the reader to treat it as the finding. The eleventh row of fourteen invites them to count. And filtering what looks unlikely starts making the board's decision for it.

#### ASK 10W

How should the over-seventies table be presented to the board?

#### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

#### TAKES AS READ

the board sees the trial's subgroup tables in closed session

#### VERDICT 86W

A single striking slice on its own page invites the reader to treat it as the finding. The same slice as the eleventh row of 14 invites them to count, which is the right reaction. Presenting all of them costs nothing and removes the choice of what to emphasise from whoever assembles the pack. Leaving it out is worse than either: the board is the body entitled to see everything, and a pack that filters what looks unlikely has started making the board's decision for it.

#### TAKEAWAY 13W

A subgroup table is reported with the number of looks that produced it.

#### END OF THE DAY 17W

A nominally striking subgroup result must be judged inside the family of subgroup analyses that produced it.

## What would we lose by stopping?

### PLAN CARD 35W

Thursday, one day out. The pack is printed, the file is locked, and tomorrow the board sits for four hours to recommend one of three things. Today you put together what it has to weigh.

### BRIEFING 11W

The board meets tomorrow, and after that the decisions are made.

### WHAT YOU DECIDE 15W

Prepare the case the board will decide on, and be honest about what stopping costs.

## 14.1 Three things the board can say

SAFETY MONITORING · MONITORING BOARD ROOM · A ROOM · PROTOCOL

### SCENE 38W

Vogt's board meets behind a closed door in 40 minutes. The adjudicated effect is modest, the stopping boundary is not crossed, and four liver cases remain unexplained. Three recommendations are printed on separate cards in front of her.

### GUIDE 57W

Three recommendations and four costs, and none of the choices is free. Pair them by asking what each one gives up: precision, a real benefit, another year of exposure, or comparability with the registered plan. Early stops overstate effects. And even changing what gets reported has a cost, unless the change is recorded before it is used.

### ASK 12W

Match each recommendation to what the board gives up by making it.

### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

### TAKES AS READ

a monitoring board recommends and the sponsor decides · sample size and statistical power — taken as read

### VERDICT 85W

None of the board's three choices is free. Stopping for benefit before the boundary is crossed risks exaggerating an effect and loses future safety follow-up. Stopping for futility can discard a modest benefit that more events might confirm. Continuing preserves the planned evidence but keeps people exposed to the arm doing worse so far. Even changing what gets reported has a cost. The registered plan is part of why readers can trust the result, so any change has to be recorded before it is used.

### TAKEAWAY 12W

Each way of ending a trial buys something and gives something up.

## 14.2 What can 65% of the information actually tell us?

STATISTICAL ANALYSIS · STATISTICS & ANALYSIS · A PERSON · BALLPARK

### SCENE 54W

Feldman has 246 of the planned 380 events and an interim z-score of 1.42. Balogun asks for the probability that the final analysis will be positive. Feldman writes two different answers on the board because the forecast changes with what you assume about the treatment effect in the events that have not happened yet.

### GUIDE 57W

Work out the one quantity the data determine without an extra assumption: the information fraction, events in hand over events planned. Then ask what is still missing before anyone can quote conditional power. The interim z-score describes what has happened. A forecast of the remaining 35% also needs an assumption about the effect in the future data.

### ASK 18W

How far through its planned information is the trial, and can that fraction by itself determine conditional power?

### BEHIND THE BUTTON 155W

What the information fraction is. Events observed over events planned. It is how far through an event-driven trial actually is, and it is not the same as how far through its calendar or its recruitment — 246 of 380 events is about two thirds of the information, whatever the enrolment log says.

What conditional power is. It is a model-based probability that the final analysis will cross a specified boundary, conditional on the data so far and on an explicit assumption about the treatment effect in the information still to come. Assuming the original design effect and assuming the current observed trend are different forecasts.

Why the forecast cannot replace the stopping rule. A high conditional power under one assumption is not the same thing as crossing a prespecified efficacy boundary. The board can use forecasts to understand consequences of continuing, but the decision rule and its operating characteristics were fixed before these data existed.

### TAKES AS READ

the trial has collected about two thirds of its planned events · sample size and statistical power — taken as read

### VERDICT 102W

The information fraction is fixed by the observed event count in this simplified design:  $246/380 \approx 0.65$ . The interim z-score of 1.42 is also observed. But conditional power is not a third observation hidden inside those two numbers. To predict the final result, the calculation must assume what treatment effect will govern the remaining events—for example, the original design alternative, the current estimated effect, or some more conservative value. Those assumptions can give materially different forecasts. The lesson is the distinction the board needs most

today: 0.65 is measurement-derived progress; conditional power is a scenario.  
Neither is the prespecified stopping boundary itself.

**TAKEAWAY 19W**

Information fraction is observed; conditional power is a forecast that changes with the assumption used for the future data.

## 14.3 What a stop obliges

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · SEQUENCE

### SCENE 35W

Umeh has the registry entry, protocol and publication policy open before the board call. One participant is due an infusion on Monday. Doyle asks what happens, in what order, if the board recommends stopping tomorrow.

### GUIDE 54W

All four steps will happen, so ask who learns what, and when. The board recommends and the sponsor decides. Somebody is due an infusion on Monday. A registered trial that quietly disappears is how the literature acquires missing arms. And a trial stopped early still owes its results, with the shortened follow-up stated plainly.

### ASK 11W

Order what has to happen if the board recommends stopping tomorrow.

### BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

### TAKES AS READ

a trial's commitments continue after recruitment ends · allocation concealment: protecting the moment of entry — taken as read

### VERDICT 84W

A stop is an event with a sequence, and the sequence is about who learns what and when. The sponsor decides, because the board only recommends. Sites and participants come next and quickly: somebody is due an infusion on Monday and is entitled to know. The public record is corrected because a registered trial that quietly disappears is how the literature acquires missing arms. And reporting is not optional — a trial stopped early still owes its results, with the shortened follow-up stated plainly.

### TAKEAWAY 13W

Ending recruitment does not end the trial's obligations to the people in it.

END OF THE DAY 17W

Stopping is a decision about what will never be known, as much as about what is known.

## Release the pack

### PLAN CARD 32W

Friday, board day. The adjudicated effect is promising but has not crossed the line the plan set. Today you decide what each number may claim, then you sign The Monitoring Board Pack.

### BRIEFING 18W

The board meets this afternoon, and the score that looks best is the one the leak can reach.

### WHAT YOU DECIDE 15W

Assemble and sign the monitoring-board pack without turning a promising interim signal into a conclusion.

## 15.1 One relative risk, many absolute effects

REGULATORY & REPORTING · REGULATORY & REGISTRY · A ROOM · DEGENERACY

### SCENE 39W

2 lines for the board chair's closed briefing describe the same CLARION-3 result. One says risk fell about 22%; the other says roughly 4 events were prevented per hundred people. Umeh puts both arm rates on a two-control display.

### GUIDE 56W

Two controls slide the two arm risks along a family of pairs that all give the same relative result. Find out how wide that family is before choosing a line for the briefing. Then bring in the second measurement, which fixes one arm rather than the ratio, and watch the family collapse to a single pair.

### ASK 18W

How many pairs of arm risks fit the relative result, and what second measurement identifies the CLARION-3 pair?

### BEHIND THE BUTTON 147W

Why a relative risk does not pin the arms. A 22% reduction is a ratio. Twenty per cent against sixteen has it; two per cent against one and a half has it too. The ratio is the same and the number of people helped differs by an order of magnitude.

What the absolute figure adds. Four events prevented per hundred people is a statement about how common the outcome is, which the ratio deliberately hides. Both sentences are true of the same result, and only one of them tells the chair what the trial would do for a patient.

Why boards are shown both. A relative figure travels well between populations and exaggerates importance in rare outcomes. An absolute figure is honest about scale and does not transfer to a population with a different baseline. Presenting one alone is the most common way a true result misleads.

### TAKES AS READ

relative risk is treatment risk divided by control risk · absolute risk reduction is control risk minus treatment risk

### VERDICT 79W

A ratio does not carry the baseline rate with it. Treatment and control risks of 3.9 and 5% give almost the same relative reduction as 14.7 and 18.9%. The absolute benefits are very different. That is why the relative result forms a whole line of possible pairs on the display. Adding the measured absolute reduction gives a second line. Their intersection recovers the actual arm rates and shows why an absolute statement carries information the relative headline leaves out.

### TAKEAWAY 15W

A relative effect does not determine the absolute benefit until the baseline risk is known.

## 15.2 What can be claimed

ENDPOINTS & MEASUREMENT · ADJUDICATION ROOM · A ROOM · CHOICE

### SCENE 30W

Ansah is asked to write one sentence about the adjudicated endpoint for the board's minutes. He has the effect, the interval, and the fact that the trial has not finished.

### GUIDE 53W

Four sentences for the minutes. Ask of each what it claims beyond the estimate and its interval. One refuses to say anything until the target is reached, which throws away information already collected. One imports a fragile subgroup. One argues the trial should have stopped, which is the question the boundary already answered.

### ASK 8W

Which sentence does CLARION-3's adjudicated endpoint currently support?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

the interval around an estimate says what the data are compatible with

### VERDICT 81W

An interim estimate is evidence, but it is not the final word. The point estimate says where the observed difference sits, and the interval shows the range still compatible with the data. More events can move both. Saying there is no effect until the target event count is reached throws away information already collected. Claiming the effect is specific to older participants imports a fragile subgroup result. Claiming the trial should have stopped ignores the stopping boundary the board actually used.

### TAKEAWAY 13W

An honest claim carries its uncertainty and the conditions it was measured under.

## 15.3 Sign the board pack

REGULATORY & REPORTING · REGULATORY & REGISTRY · A PERSON · CHOICE

### SCENE 45W

It is 2:42. Vogt's board meets at three. Four versions of the executive page are on Umeh's screen. They contain the same numbers and make different claims from them. Once you sign, Petrescu's closed arm-by-arm tables are attached and the pack leaves the trial team.

### GUIDE 92W

Do not choose the version with the most cautious adjective or the most dramatic number. Trace each sentence back to the evidence chain. The adjudicated endpoint is promising and has not crossed the prespecified efficacy boundary. The symptom score has a demonstrated path for treatment knowledge to enter. The liver cases are a real safety signal whose arm pattern belongs to the board's protected review. The over-seventies row is one of fourteen subgroup looks. The signed page must preserve all four facts without deciding by wording what the board exists to decide.

### ASK 12W

Which executive page can you sign and release to the monitoring board?

### BEHIND THE BUTTON 166W

What the signature means. It does not certify that the drug works or that the trial should continue. It certifies that the board can see the prespecified question, the current estimate and uncertainty, the relevant limitations, and the unresolved safety evidence without one being hidden inside another.

Why the tempting versions fail. A large subjective effect is not independent of a blinding failure that reaches the scorer. A nominal subgroup result is not a new primary question. Four liver cases do not prove harm without the exposure and arm pattern, but they cannot be buried behind efficacy either. And a result can be statistically promising without satisfying the rule written for stopping early.

What happens next. The independent board sees the protected arm-specific data and makes its recommendation. The sponsor then acts on that recommendation under the protocol. If the trial continues, sites preserve the blind and participants receive whatever safety or consent information the protocol and ethics review require without publishing protected interim efficacy data.

### TAKES AS READ

the trial team remains blinded to comparative interim efficacy data · the independent board receives the protected arm-specific safety and efficacy tables

### VERDICT 133W

Every sentence on the qualified page has a different evidential status. The adjudicated endpoint is the prespecified efficacy measurement least exposed to the blinding failure and is promising without crossing the early-stopping boundary. The symptom score is not erased, but its interpretation is limited because treatment knowledge can reach its scorer. Four liver cases require arm-specific safety review

rather than an automatic causal conclusion. The over-seventies result is a nominally striking row inside a family of subgroup analyses, not a new primary claim. Signing anything stronger converts one of those limitations or assumptions into a fact. Signing anything weaker hides evidence the board is supposed to weigh. This is the job the fifteen days built toward: release a record strong enough for a consequential decision without making the decision by misdescribing the evidence.

**TAKEAWAY 18W**

The pack is ready only when every claim is traceable to evidence and every unresolved question stays visible.

**END OF THE DAY 19W**

The final job is to hand the board a decision-ready record that keeps measurements, assumptions, limitations and recommendations separate.

## THE CLOSING CARD

At three twenty-seven Vogt closes the pack you signed. The board recommends continuing to the planned event target, keeping the investigator-rated symptom score out of the headline efficacy claim and putting the four liver cases under formal safety follow-up. Eleven months later CLARION-3 locks on 383 adjudicated events. The final absolute risk reduction is 3.6 percentage points — about twenty-eight people treated for one event avoided — with a confidence interval that excludes zero but is wider than the team had hoped.

What it cost: eleven more months of randomisation under genuine uncertainty; in hindsight, about half of those newly enrolled received the less effective arm. The subjective symptom result remains difficult to interpret because the blind leaked. The over-seventies signal is not replicated in two later studies. What is unfinished: long-term effects past two years, the meaning of four liver cases, and how a future trial hides an early visible treatment effect without letting that knowledge reach the outcome measurement.

The board could make its decision because the pack did not pretend those questions were settled. You separated the endpoint that survived the blinding failure from the one that did not, kept the subgroup inside the family of fourteen that produced it, and labelled a forecast as an assumption instead of a measurement. The decision was the board's. The evidence it decided from was yours.

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Card text only — no options, boards or answers. 11,512 words of card prose across 51 stops. Generated from `themes/the_trial` by `tools/make-cardbook.mjs`.